



LLM-based Academic Writing with Fine-Grained Citations

Master Thesis Presentation

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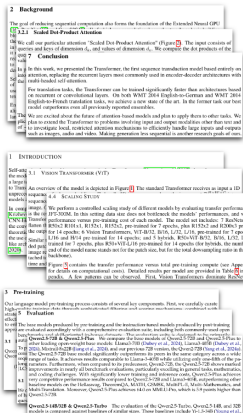


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Reference Retrieval with the Passage Retrieval Module

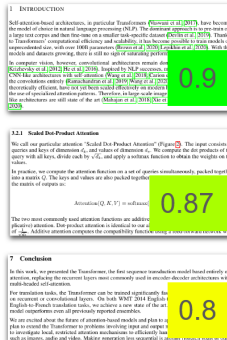
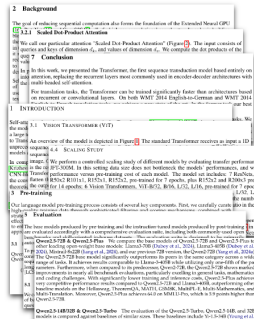


Ranking by ScholarCopilot
based on abstracts

Passages from documents
retrieved by ScholarCopilot



Reference Retrieval with the Passage Retrieval Module (continued)



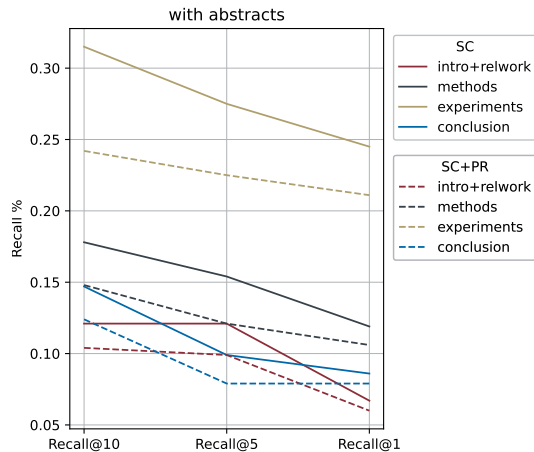
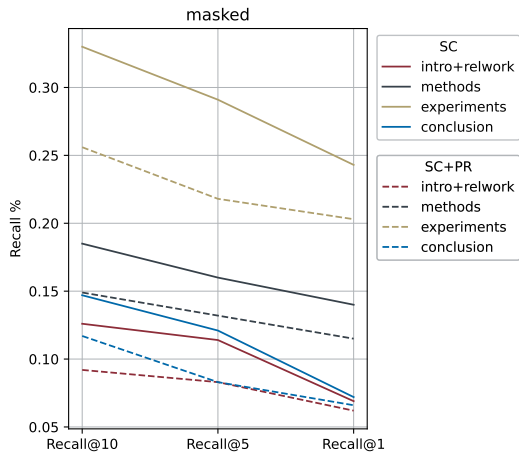
Passages from documents
retrieved by ScholarCopilot

Ranking by the PR module
based on passages

Reference retrieved by the
PR module



Citation Recall Performance





Recall@ k decline towards small k

Model	Dataset	intro+relwork	methods	experiments	conclusion	Σ
SC	masked	-0.06	-0.04	-0.09	-0.07	-0.26
SC	with-abstracts	-0.05	-0.06	-0.07	-0.06	-0.24
SC+PR	masked	-0.03	-0.03	-0.05	-0.05	-0.16
SC+PR	with-abstracts	-0.04	-0.04	-0.03	-0.04	-0.15

Table: Cumulative Recall@ k difference from $k = 10$ over $k = 5$ to $k = 1$.



Why is the recall for experiments so much higher?

	intro+relwork	methods	experiments	conclusion
last token in abstract	0.08	0.12	0.22	0.11

Table: Fraction of cases where the last token before the citation appears in the abstract of the cited document.

Correct Retrieval Overlap

Dataset	Metric	intro+relwork	methods	experiments	conclusion	μ
masked	Recall@10	0.91	0.97	0.98	0.91	0.943
masked	Recall@5	0.94	0.94	0.96	0.94	0.945
masked	Recall@1	0.79	0.89	0.88	0.88	0.86
with-abstracts	Recall@10	0.9	0.93	0.96	0.95	0.935
with-abstracts	Recall@5	0.86	0.97	0.95	0.89	0.918
with-abstracts	Recall@1	0.87	0.87	0.93	0.84	0.878

Table: Fraction of correct retrievals by ScholarCopilot given correct retrieval by the PR module.



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